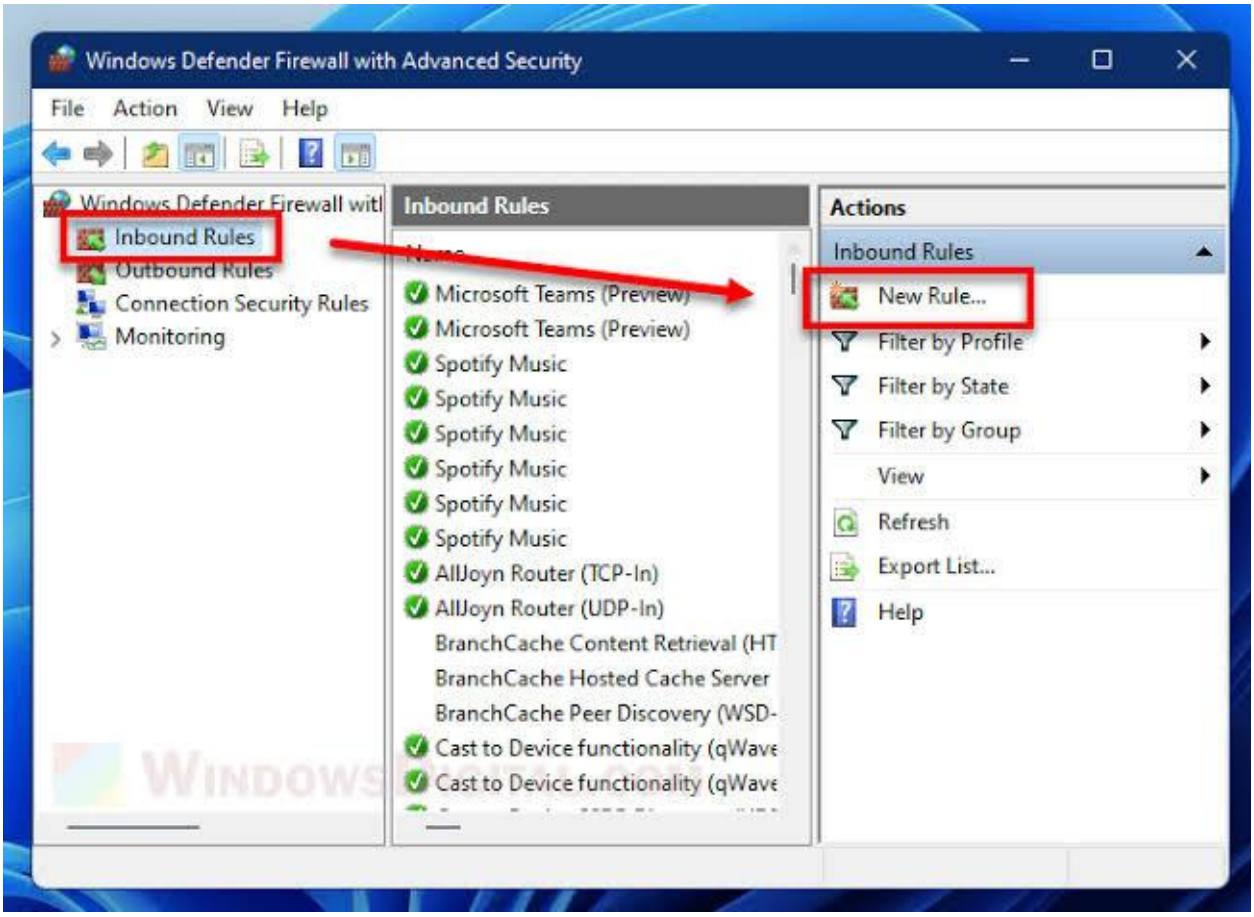
Assignment 20

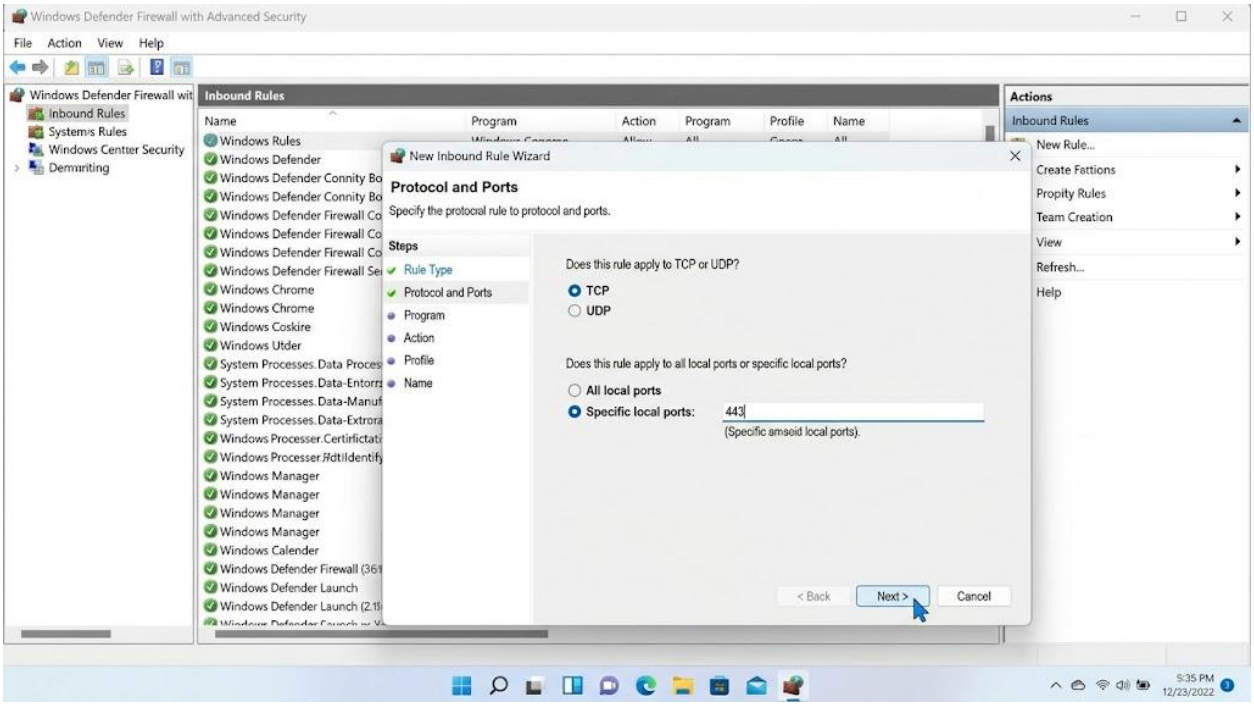
Firewall Basic Theory

1. **Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.**



1. **Open Advanced Firewall Settings**
2. **Initiate a New Outbound Rule**
3. **Select Rule Type**
4. **Locate Spotify Executable**
5. **Configure Action to Block**
6. **Apply Network Profiles**
7. **Name and Save the Rule**

2. **Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).**



3. **Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.**

❖ Steps to Block an IP Address in Windows Defender Firewall.

- Open **Windows Defender Firewall** with Advanced Security from the Windows Start menu.
- Click **Inbound Rules** on the left side.
- Click **New Rule** on the right side.
- Select **Custom** and click Next.
- Select **All programs** and click Next.
- In the **Protocol** and **Ports** section, keep the **default settings** and click Next.
- In the **Scope tab**, under Which remote IP addresses does this rule apply to?, select These IP addresses.
- Click **Add**, enter **123.45.67.89**, and click OK.
- Click **Next**, select Block the connection, and click Next.
- Select all three profiles: Domain, Private, and Public. Click Next.
- Enter the rule name as Block Specific IP and click Finish.

- Result: Windows Defender Firewall will block incoming traffic from 123.45.67.89.

4. **Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each**.

Difference	Hardware Firewall	Software Firewall
1. Location	Installed as a physical device in the network.	Installed as software on a computer.
2. Protection	Protects multiple devices in a network.	Protects an individual computer or device.
3. Cost	Usually more expensive.	Usually cheaper or built into the operating system.

Use cases:

1. **Hardware Firewall:** Used in large offices to protect the entire company network from hackers and unwanted traffic.

2. **Software Firewall:** Windows Defender Firewall is used on a personal laptop to block unauthorized connections and protect the device.